microCT imaging of threespine stickleback

This manuscript (permalink) was automatically generated from habi/sticklebacks-manuscript@b94bcb6 on January 15, 2026.

Authors

- **David Haberthür**
  0000-0003-3388-9187 · habi · @habi@mastodon.social
  microCT research group, Institute of Anatomy, University of Bern, Baltzerstrasse 2, 3012 Bern, Switzerland

- **Ben Sulser**

- **Sheila Christen**

- **Katie Peichel**

- **Ruslan Hlushchuk** ✉
  0000-0001-2345-6789 · RuslanHlushchuk
  microCT research group, Institute of Anatomy, University of Bern, Baltzerstrasse 2, 3012 Bern, Switzerland

✉ — Correspondence possible via GitHub Issues or email to Ruslan Hlushchuk <ruslan.hlushchuk@unibe.ch>.

Abstract

Introduction

- Embedded into Alaska Stickleback Restoration Project, Genomics axis where Katie Peichel, Ben Sulser and Sheila Christen are affiliated.

Materials & Methods

Sample preparation

microtomographic imaging

- Sample holder generated with OpenSCAD, visible at https://github.com/TomoGraphics/Hol3Drs/blob/master/STL/Stickleback.Multiple.stl

```
solid cube_corner
  facet normal 0.0 -1.0 0.0
    outer loop
      vertex 0.0 0.0 0.0
      vertex 1.0 0.0 0.0
      vertex 0.0 0.0 1.0
    endloop
  endfacet
  facet normal 0.0 0.0 -1.0
    outer loop
      vertex 0.0 0.0 0.0
      vertex 0.0 1.0 0.0
      vertex 1.0 0.0 0.0
    endloop
  endfacet
  facet normal -1.0 0.0 0.0
    outer loop
      vertex 0.0 0.0 0.0
      vertex 0.0 0.0 1.0
      vertex 0.0 1.0 0.0
    endloop
  endfacet
  facet normal 0.577 0.577 0.577
    outer loop
      vertex 1.0 0.0 0.0
      vertex 0.0 1.0 0.0
      vertex 0.0 0.0 1.0
    endloop
  endfacet
endsolid
```

Data analysis

- Jupyter notebooks

Results

Discussion

Conclusion

Author Contributions

Contributor Roles Taxonomy (CRediT), as defined in [1]:

- Data curation: David Haberthür
- Formal analysis: David Haberthür
- Investigation: David Haberthür
- Methodology: David Haberthür
- Project administration: David Haberthür
- Software: David Haberthür
- Validation: David Haberthür
- Visualization: David Haberthür
- Writing – original draft: David Haberthür
- Writing – review & editing: David Haberthür

Competing Interest

Author	Competing Interests	Last Reviewed
David Haberthür	None	2026-01-14
Ben Sulser		
Sheila Christen		
Katie Peichel		
Ruslan Hlushchuk		

Acknowledgements

We are grateful to the Microscopy Imaging Center of the University of Bern for their infrastructural support. We also thank the `manubot` project [2] for facilitating collaborative writing of this manuscript.

References

1. **ANSI/NISO Z39.104-2022, CRediT, Contributor Roles Taxonomy** *NISO* https://doi.org/gqx265
DOI: 10.3789/ansi.niso.z39.104-2022

2. **Open collaborative writing with Manubot**
Daniel S Himmelstein, Vincent Rubinetti, David R Slochower, Dongbo Hu, Venkat S Malladi, Casey S Greene, Anthony Gitter
PLOS Computational Biology (2019-06-24) https://doi.org/c7np
DOI: 10.1371/journal.pcbi.1007128 · PMID: 31233491 · PMCID: PMC6611653